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SYSTEMATIC REVIEW

Does Resisted Sprint Training Improve the Sprint Performance of Field-Based Invasion Team Sport Players? A Systematic Review and Meta-analysis

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Abstract

Background Developing the sprint performance of field-based invasion team sport (FITS) players is considered an essential training goal for FITS coaching practitioners, and thus numerous training methods are employed to elicit improvements. Although interest in resisted sprint training (RST) has grown considerably in recent times, there remains a lack of clarity around its utility in FITS, particularly regarding the use and effectiveness of heavier RST loads.

Objectives The aims of this review were to (1) compare RST to unresisted sprinting, (2) examine if RST can improve sprint performance and (3) investigate if external load and the method of load prescription influence the impact of RST in FITS players.

Methods The systematic review and meta-analysis were conducted in compliance with the Preferred Reporting Items for Systematic Review and Meta-Analyses (PRISMA) guidelines. The search strategy included terms for RST, RST modalities and FITS, and was applied to PubMed, SPORTDiscus, Web of Science and OpenGrey databases. Methodological quality and risk of bias associated with each study were assessed using the Physiotherapy Evidence Database scale (PEDro) and Cochrane Risk of Bias assessment tool respectively.

Results Twenty-one studies met the inclusion criteria for this review and were included in the final analysis. The primary between-group analysis revealed no differences between RST and unresisted sprinting for developing the early acceleration, late acceleration and maximum velocity sprint phases. Secondly, a within-group analysis found significant improvements for resisted sprint training in the early acceleration (standardised mean difference [SMD] − 0.80) and late acceleration (SMD − 0.28) sprint phases, with no change detected for the maximum velocity phase. Finally, significant moderate improvements were found for light (SMD − 0.69) and very heavy (SMD − 1.01) loads during early acceleration.

Conclusions Resisted sprint training achieved similar improvements in sprint performance to those found for unresisted sprinting during the acceleration and maximum velocity sprint phases. Within-group findings show RST is an effective method for improving early acceleration and late acceleration performance of FITS players. Finally, a subgroup analysis supports the use of light and very heavy loads for increasing early acceleration performance, while also highlighting greater benefits associated with using the percentage velocity decrement loading method.

Clinical Trial Registration Open Science Framework, https://osf.io/thms7/.

1 Introduction

Field-based invasion team sports (FITS) [e.g., soccer, rugby, Australian Rules football] are characterised by competitive match play between two sets of players who perform an array of complex skilful actions at varying physical intensities, over an extended period of time (typically 60–90 min). These physical intensities are primarily dictated by the sport's intermittent running demands, which consist of maximal or near-maximal efforts interspersed with low-to-moderate intensity recovery periods [1]. Of these demands, sprint performance is considered a determining factor in the outcome of games [2, 3], and as such is associated with the successful completion of key attacking and defensive performance indicators [4–6]. In fact, rapid linear accelerations have been shown to regularly precede or be directly involved in the creation of scoring opportunities [2]. Therefore, enhancing the sprinting ability of players is a key training goal for FITS coaches.

Extended author information available on the last page of the article

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Key Points

No differences were found between resisted sprint training and unresisted sprinting for increasing the sprint performance of field-based invasion team sport players.

Resisted sprint training is an effective method to increase the acceleration performance of field-based invasion team sport players with early acceleration demonstrating a larger effect size (standardised mean difference – 0.80) than late acceleration (standardised mean difference – 0.28).

Light (2.5–10% velocity decrement) and very heavy (≥ 50% velocity decrement) loads were found effective for improving early acceleration performance. This result may suggest that performing resisted sprint training with loads spanning large ranges of the percentage velocity decrement loading method will invariably elicit improvements to early acceleration in comparison to the late acceleration and maximum velocity phases where there may be a requirement for greater velocity specificity.

Training methods used to enhance sprint performance typically aim to improve an individual's ability to produce force [7, 8], and to increase efficiency in the use and application of this force [9, 10]. Resistance training methods such as strength (high force), power/ballistic and complex training, along with bodyweight plyometric training (reactive strength) have all been shown to improve (0.43–8.8%) sprint performance [8, 11–14]. However, given that sprinting is a cyclical unilateral movement primarily concerned with rapid whole-body horizontal displacement [15], it is evident that exercises used within these traditional methods (e.g. back squat, jump squats) may lack the specificity required to elicit more substantial increases in sprint performance [8, 16]. The principle of specificity states that the training effect that occurs in response to an exercise overload is specific to the way in which the load was applied [17]. Considering this, free or unresisted sprinting (URS) is commonly regarded as the primary training method for developing sprint performance, although other methods exist that allow the retention of a higher degree of task specificity when compared with traditional resistance methods [18, 19]. Resisted sprint training (RST) applies an external load/resistance to the sprinting movement [20], potentially allowing for the maintenance of sprint specific mechanical properties (e.g. motor pattern and contraction type) while developing a player's capacity to produce greater force [21–24]. In accordance with the principle of dynamic correspondence, this may result in a greater transfer effect from training to performance [15, 25]. Thus, RST may represent a more specific, and potentially more effective method for enhancing sprint performance than traditional training methods (e.g. strength training and plyometrics [13]).

A multitude of interventions and thus numerous reviews [13, 18, 19, 26–29] have investigated and discussed the effectiveness and prescription of RST for improving sprint performance. Of these reviews, two were narrative [18, 27] and five reported systematic search strategies and meta-analysis [13, 19, 26, 28, 29]. In their narrative review, Cahill et al. [27] highlighted the need to consider multiple factors when prescribing RST using sleds, which included differences in athletes' physical characteristics (e.g. sex, relative strength, body composition) and surface type. In turn, the authors recommended using the percentage velocity decrement loading method (%V_{dec}) over the percentage body mass (%BM) method to help cater for these differences, while also providing loading recommendations for targeting specific stimuli, for example, > 65% V_{dec} loads reflecting strength-speed. Reviews [26, 28, 29] that focused exclusively on RST reported that the greatest improvements in sprint performance were observed during early acceleration (EA), with only small effects being generally detected during the maximum velocity (MV) phase. Furthermore, these reviews concluded that RST and URS appear capable of eliciting similar improvements to sprint performance, particularly over the acceleration phase. Despite the significant contribution of these reviews [13, 18, 19, 26–29], the interest and popularity of RST have continued to grow among coaches and researchers, which in turn highlights the need to provide greater clarity regarding its specific utility among FITS players.

Previous systematic reviews have examined mixed populations consisting of sprinters, recreational athletes, FITS players and other non-field-based team sport players [26, 28], making it difficult for FITS coaches/players to interpret and apply study findings. Research shows that FITS players achieve lower maximum sprinting velocities at shorter distances than sprinters [30–32], while also displaying differences in force-producing capabilities (sprint specific) and sprint technique [31, 33]. Furthermore, unlike individual sports, the more multifaceted nature of FITS [1] results in the need for players to concurrently develop multiple physical qualities (e.g. speed, strength, endurance) alongside technical competence [34, 35], which may impact training adaptations [36] or the overall effect of training interventions (e.g. RST). Therefore, it is apparent that there is a need to investigate the effectiveness and application of RST with a specific emphasis on the FITS player [18]. Additionally, since the published reviews of Petrakos et al. [28] and Alcaraz et al. [26], renewed interest in RST with heavier loads has seen five additional studies published in this area [37–41], with a further eight interventions utilising light and moderate loads [42–49]. Indeed, the recent narrative

review by Zabaloy et al. [18] questions the need to use heavy (30–< 50% V_{dec}), and very heavy (≥ 50% V_{dec}) loads, with little evidence suggesting they provide any benefit above those found with lighter loads. The authors proposed that because of factors such as disruptions in sprint technique, use of heavy and very heavy loading magnitudes should be considered a tertiary sprint training method and be employed using short sprint distances and lower volumes. Therefore, given that research suggests that sprint phase adaptations are velocity specific or a function of RST load and distance [21, 28, 37, 38, 50], it is important for FITS coaches and players to understand how a broad range of loads impact sprint performance. Therefore, this review aims to (1) compare RST to URS, (2) examine if RST can improve sprint performance and (3) investigate if the external load and the method of load prescription influences the impact of RST in FITS players.

2 Methods

A systematic review and meta-analysis was conducted in compliance with the Preferred Reporting Items for Systematic Review and Meta-analyses (PRISMA) guidelines [51]. The review protocol was registered on the Open Science Framework database, https://osf.io/thms7/.

2.1 Literature Search

The following databases were used to identify studies for this review: PubMed, SPORTDiscus and Web of Science. A grey literature search was also conducted using the OpenGrey database. Searches were performed separately by two reviewers (CW and SG) and results were limited to studies/articles dated up to and including the 30 November, 2022. A computerised search of the literature was performed using a search strategy (Fig. S1 of the Electronic Supplementary Material [ESM]) consisting of specific Boolean phrases and keywords such as "sprint" AND "resisted sprint training" OR "sled" OR "weighted vest". Results were further limited by language (English).

2.2 Inclusion Criteria

The inclusion criteria dictated that studies were required to (1) include an experimental intervention period of at least 4 weeks during which at least one study group used a resisted sprinting device and this was the primary difference between groups, (2) include participants between the ages of 16 and 35 years who were currently playing competitive FITS, (3) present necessary descriptive information about the participants (e.g. age, body mass, training status), the resistance applied to each sprint, and the distances and volume per training session during the intervention, (4) include either a non-exposed control group or a comparison group (Table S1 of the ESM) and finally, (5) include pre- and post-intervention sprint test times measured by an automated electronic device, for example, timing gates or radar gun.

Studies that provided pre- and post-test results of sprint performance (over given distances) in the form of average speed (meters per second [m/s]) as opposed to time (seconds [s]) were converted to time by dividing the test distance by the average speed. For the purpose of conducting the primary analysis of the review, sprint performance was subdivided into three specific phases that were adapted from two previous systematic reviews [26, 28], each defined in Table 1.

Studies were included in the review if at least one sprint phase analysis could be conducted using the testing distances and split times provided. Where sprint times for two or more phases were provided, time for additional sprint phases was calculated through the subtraction or summation of mean data.

2.3 Data Extraction

The title and abstract of the studies identified in the initial database search were independently screened by two reviewers (CW and SG) using the inclusion criteria above. The full text of all articles that passed the title and abstract screening process were then independently screened by the same reviewers. A third reviewer (DTK) was available to resolve any disagreements regarding the inclusion and exclusion of studies but was not required. The study process is illustrated in Fig. 1.

Relevant data were extracted from all included studies and input into Microsoft Excel by reviewer one (CW). Reviewer two (SG) independently cross-checked the accuracy of the data per cell. A third reviewer (DTK) was available to cross-check and resolve any discrepancies but was not required.

Table 1 Sprint phases [26, 28]

Phase	Definition
Early acceleration	Time recorded to sprint a distance from 0 to ≤ 10 m
Late acceleration	Time recorded to sprint a distance from 10 to ≤ 20 m
Maximum velocity	Time recorded to sprint a distance ≥ 10 m at maximum intensity using a run-in distance of ≥ 20 m

2.4 Statistical Analyses

Review Manager Software (RevMan 5.4; Cochrane Collaboration, Oxford, UK) was used to conduct a random-effects meta-analysis [52]. Effect size is presented as standardised mean difference (SMD) with 95% confidence intervals (CIs) and reported according to the following effect size threshold values: trivial (< 0.2), small (0.2–0.59), moderate (0.60–1.19), large (1.2–1.99), very large (≥ 2) or extremely large (≥ 4) [53]. The SMD used in this meta-analysis is the effect size known as Hedges' g [54] and is calculated as the difference in means divided by the pooled standard deviation and multiplied by the correction factor to yield an unbiased

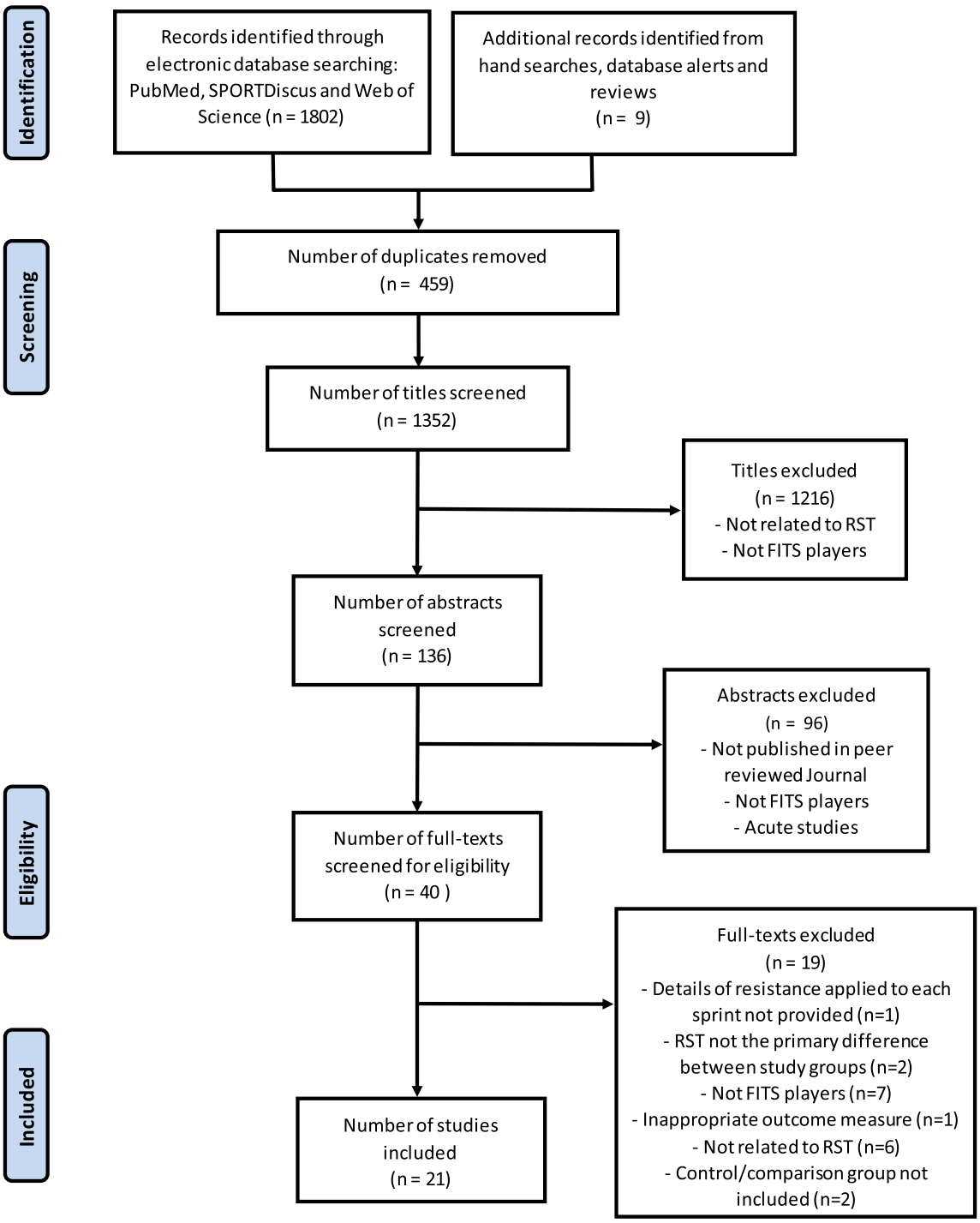

Fig. 1 PRISMA flow diagram representing the study selection process. *FITS* field-based invasion team sport, *RST* resisted sprint training

effect estimate [55, 56]. A p-value of < 0.05 was considered statistically significant.

To assess the effects of RST on sprint performance a between-group (experimental group [RST] comparison to control group [URS]) analysis was conducted, whereby the difference in sprint times (post–pre) and standard deviations were calculated for each group and input to RevMan along with the number of participants in both groups, allowing calculation of the overall SMD. A secondary within-group analysis was conducted to further examine the effectiveness of RST on sprint performance, with pre- and post-intervention sprint times, standard deviations and the number of participants (pre and post) from each RST group input to RevMan in order to generate an overall SMD.

Finally, a subgroup analysis was conducted to assess the impact of the following moderator variables on sprint performance: (1) external load (%V_{dec} studies only) and (2) loading method (%BM vs %V_{dec}). For a subgroup analysis of RST load, the same load categories established by Zabaloy et al. [18] were used. These were light (low) (2.5–10% V_{dec}), moderate (> 10–< 30% V_{dec}), heavy (30–< 50% V_{dec}), and very heavy (≥ 50% V_{dec}) categories.

2.5 Study Quality, Risk of Bias, and Heterogeneity

Methodological quality of each study was independently assessed by two reviewers (CW and DTK) using the Physiotherapy Evidence Database scale (PEDro) [57, 58]. The scale scores ten items, with each allocated a score of 1 (present) or 0 (absent), and a total score out of a maximum of 10 is obtained. Risk of bias was independently assessed by two reviewers (CW and DTK) using the Cochrane Risk of Bias assessment tool [59]. A third independent reviewer (COC) was available to resolve any disagreements in these judgements in accordance with Cochrane Collaboration guidelines [60], however, they were not required. Heterogeneity among studies was assessed using I^2 statistics with values of > 50% and > 75% considered as moderate and high, respectively [61].

3 Results

Twenty-one studies [21, 37–49, 62–68] met the inclusion criteria. In total, this sample included 31 study groups of which 12 studies included a non-exposed control group [21, 37, 39, 43, 44, 47–49, 62, 66–68], with the remaining studies eligible for inclusion owing to the presence of a comparison group.

3.1 Participant Characteristics

The age of the participants ranged from 16.6 (± 0.6) to 27.1 (± 4.8) years and training level ranged from trained to highly trained. Only three studies included female players [38, 39, 49]. Soccer ($n = 14$), rugby ($n = 7$), lacrosse ($n = 2$), field hockey ($n = 1$) and Australian Rules football ($n = 2$) were the recruited sports across the 21 studies (Table S2 of the ESM).

3.2 Intervention Characteristics

Resisted sprint training modalities included the 1080 Sprint (portable robotic resistance device [1080 Sprint™; 1080 Motion, Lidingö, Sweden]), Vertimax (platform with elastic cords and sheaves [Vertimax®, Model V8, Genetic Potential, Tampa, FL, USA]), The Trainer (portable resistance device containing an in-built load calibration scale (Perform Better, Cranston, RI, USA), weighted vests, and weighted sleds. Studies were carried out in-season ($n = 13$), pre-season ($n = 5$) and during off-season ($n = 3$). Interventions lasted between 4 and 11 weeks (7.3 ± 1.7) and the total number of sessions completed ranged from 9 to 16 (13 ± 2.4). Resisted sprint training repetition distance was between 5 and 55 m, with total RST session volume and intervention volume that ranged from 30 to 455 m and 380 m to 4800 m, respectively. A total of 13 studies used %BM to prescribe loads that ranged between 10 and 80% BM [21, 42, 43, 45–49, 63, 64, 66, 68], while eight studies used %V_{dec} with a loading range of 10–80% V_{dec} [37–41, 44, 65, 67] (Tables S3 and S4 of the ESM).

3.3 Study Quality, Risk of Bias and Heterogeneity

The mean score for methodological quality using the PEDro scale was 6, with a score of 5–6 considered to have 'fair' to 'good' reliability [69]. A high risk of bias was concluded for both domains within the Cochrane Risk of Bias assessment tool [59] relevant to 'blinding'. However, while it is typically not feasible to achieve blinding in supervised training interventions [70], the possible impact of performance and detection bias among studies must be acknowledged. The remaining domains produced a low and unclear risk of bias. The extent of overall heterogeneity throughout all analyses was considered low; however, some isolated cases of moderate heterogeneity were detected during subgroup analyses.

3.4 Early Acceleration Phase

Sixteen experimental study groups ($n = 171$) from 11 studies [21, 37, 39, 43, 44, 47–49, 66–68] were included in the between-group analysis to determine the effect of RST on EA phase sprint performance. Results displayed no differences between groups ($p = 0.24$; SMD − 0.13; 95% CI − 0.35 to 0.09) (Fig. 2). Twenty-nine study groups ($n = 307$) from 20 studies [21, 37–49, 63–68] were

included in the within-group analysis to determine the effect of RST on sprint time in the EA phase, with RST demonstrating a significant moderate improvement ($p <$ 0.00001; SMD − 0.80; 95% CI − 1.01 to − 0.59) [Fig. 3].

A subgroup analysis showed no statistically significant effect on EA performance due to the RST external load. The effect was greatest when using very heavy loads (SMD − 1.01) followed by light loads (SMD − 0.69). The type of loading method used had no statistically significant effect on EA performance. Using $\%V_{dec}$ (SMD − 0.84) contributed to greater effects when compared with %BM (SMD − 0.75) (Tables S5 and S6 of the ESM).

3.5 Late Acceleration Phase

Nine experimental study groups ($n = 102$) from five studies [37, 43, 44, 47, 67] were included in the between-group analysis to determine the effect of RST on LA phase sprint performance. Results displayed no differences between groups ($p = 0.92$; SMD − 0.01; 95% CI − 0.26 to 0.29) (Fig. 4). Seventeen study groups ($n = 194$) from ten studies [37, 38, 41, 43–47, 63, 67] were included in the within-group analysis to determine the effect of RST on sprint time in the late acceleration (LA) phase, with RST demonstrating a significant small improvement ($p = 0.006$; SMD − 0.28; 95% CI − 0.49 to − 0.08) [Fig. 5].

A subgroup analysis indicated no statistically significant effect on LA performance based on the RST external load. The type of loading method used had no statistically significant effect on LA performance. Using $\%V_{dec}$ (SMD − 0.27) contributed to significantly greater effects when compared with %BM (SMD − 0.30) (Tables S5 and S6 of the ESM).

3.6 Maximum Velocity Phase

Eight experimental study groups ($n = 79$) from four studies [39, 43, 49, 62] were included in the between-group analysis to determine the effect of RST on MV phase sprint performance. Results displayed no differences between groups ($p = 0.69$; SMD 0.06; 95% CI − 0.25 to 0.38) (Fig. 6). Twelve study groups ($n = 118$) from seven studies [39, 41, 43, 46, 49, 62, 63] were included in the within-group analysis to determine the effect of RST on sprint time in the MV phase. Resisted sprint training demonstrated no improvement between pre- and post-test measurements ($p = 0.12$; SMD − 0.20; 95% CI − 0.46 to − 0.06) [Fig. 7]. A subgroup analysis showed no statistically significant effect on MV performance due to the RST external load, or the type of loading method used (Tables S5 and S6 of the ESM).

4 Discussion

The purpose of this systematic review and meta-analysis was to (1) compare RST to URS, (2) examine if RST can improve sprint performance and (3) investigate if external load and the method of load prescription influences the impact of RST in FITS players. Results indicate that both URS and RST demonstrate similar effects on sprint performance during all sprint phases. A within-group analysis determined RST was most effective for improving EA,

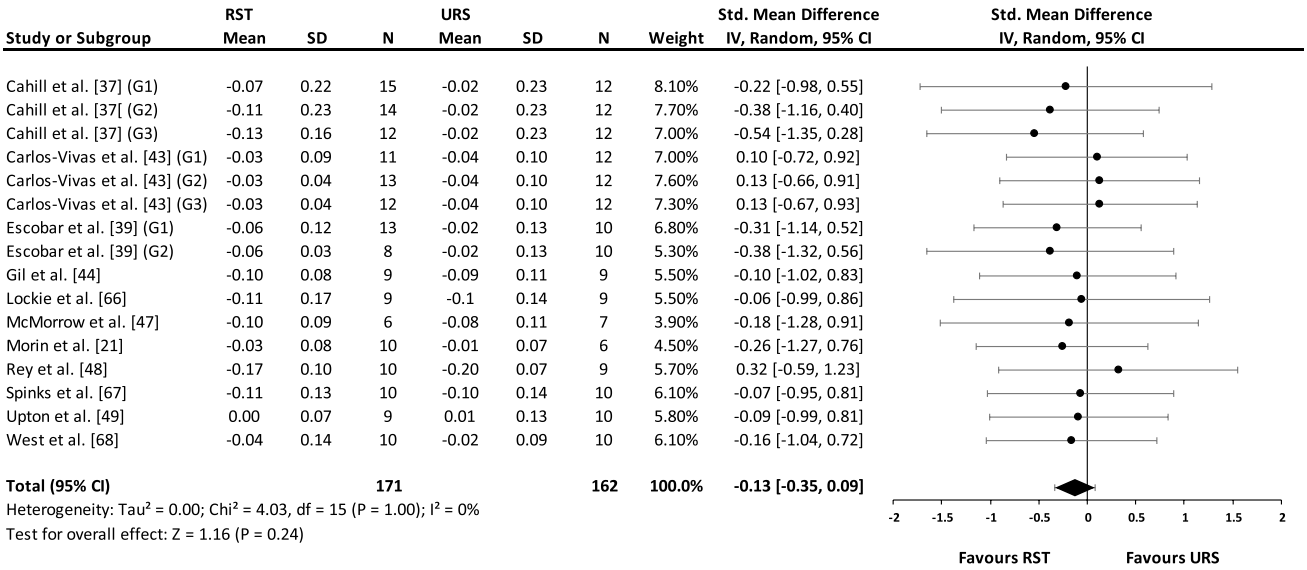

Fig. 2 Comparison of resisted sprint training (RST) and unresisted sprinting (URS) for improving early acceleration sprint performance. Circles represent the standardised mean difference (SMD) for each study group. Diamonds represent the pooled SMD for all study groups combined. *CI* confidence interval, *G1* group 1, *G2* group 2, *G3* group 3, *IV* independent variable, *SD* standard deviation

Study or Subgroup	Post Intervention Mean	SD	N	Pre Intervention Mean	SD	N	Weight	Std. Mean Difference IV, Random, 95% CI
Borges et al. [42]	1.02	0.07	9	1.03	0.03	9	3.5%	-0.18 [-1.10, 0.75]
Cahill et al. [37] (G1)	2.29	0.12	15	2.36	0.18	15	4.7%	-0.45 [-1.17, 0.28]
Cahill et al. [37] (G2)	2.34	0.14	14	2.45	0.18	14	4.4%	-0.66 [-1.43, 0.10]
Cahill et al. [37] (G3)	2.29	0.11	12	2.42	0.12	12	3.8%	-1.09 [-1.96, -0.22]
Carlos-Vivas et al. [43] (G1)	2.07	0.06	11	2.10	0.07	11	3.9%	-0.44 [-1.29, 0.41]
Carlos-Vivas et al. [43] (G2)	2.08	0.02	13	2.11	0.03	13	4.0%	-1.14 [-1.98, -0.30]
Carlos-Vivas et al. [43] (G3)	2.08	0.03	12	2.11	0.03	12	3.9%	-0.97 [-1.82, -0.11]
Cross et al. [38] (G1)	2.27	0.18	18	2.32	0.20	18	5.2%	-0.26 [-0.91, 0.40]
Cross et al. [38] (G2)	2.27	0.18	18	2.31	0.22	18	5.2%	-0.19 [-0.85, 0.46]
De Hoyo et al. [63]	1.71	0.06	13	1.72	0.05	13	4.4%	-0.18 [-0.95, 0.60]
Escobar et al. [39] (G1)	1.60	0.08	13	1.66	0.09	13	4.2%	-0.68 [-1.48, 0.11]
Escobar et al. [39] (G2)	1.45	0.02	8	1.51	0.02	8	1.7%	-2.84 [-4.33, -1.34]
Gil et al. [44]	1.69	0.10	9	1.79	0.10	9	2.7%	-1.72 [-2.85, -0.60]
Grazioli et al. [45] (G1)	1.52	0.06	8	1.60	0.05	8	2.7%	-1.37 [-2.49, -0.25]
Grazioli et al. [45] (G2)	1.55	0.09	9	1.57	0.11	9	3.5%	-0.19 [-1.12, 0.74]
Harrison & Bourke [64]	2.35	0.10	8	2.49	0.21	8	3.0%	-0.80 [-1.84, 0.23]
Kawamori et al. [65] (G1)	1.94	0.08	10	2.05	0.07	10	3.1%	-1.40 [-2.40, -0.40]
Kawamori et al. [65] (G2)	2.00	0.10	11	2.06	0.10	11	3.9%	-0.58 [-1.43, 0.28]
Lahti et al, [40]	1.33	0.02	6	1.38	0.05	6	2.2%	-1.21 [-2.49, 0.06]
Lahti et al. [41] (G1)	2.07	0.06	9	2.14	0.06	9	3.1%	-1.11 [-2.12, -0.10]
Lahti et al. [41] (G2)	2.09	0.06	9	2.15	0.08	10	3.4%	-0.80 [-1.75, 0.14]
Lockie et al. [66]	1.98	0.11	9	2.09	0.13	9	3.2%	-0.87 [-1.85, 0.11]
Loturco et al. [46]	1.64	0.06	7	1.74	0.04	7	2.1%	-1.84 [-3.16, -0.52]
McMorrow et al. [47]	1.84	0.06	6	1.94	0.07	6	2.1%	-1.42 [-2.74, -0.09]
Morin et al. [21]	1.40	0.06	10	1.43	0.04	10	3.6%	-0.56 [-1.46, 0.33]
Rey et al. [48]	1.61	0.06	10	1.78	0.07	10	2.3%	-2.50 [-3.73, -1.27]
Spinks et al. [67]	1.27	0.09	10	1.38	0.09	10	3.3%	-1.17 [-2.14, -0.21]
Upton et al. [49]	1.21	0.05	9	1.21	0.05	9	3.5%	0.00 [-0.92, 0.92]
West et al. [68]	1.70	0.10	10	1.74	0.10	10	3.7%	-0.38 [-1.27, 0.50]
Total (95% CI)			**306**			**307**	**100.0%**	**-0.80 [-1.01, -0.59]**

Heterogeneity: Tau² = 0.11; Chi² = 42.65, df = 28 (P = 0.04); I² = 34%
Test for overall effect: Z = 7.37 (P < 0.00001)

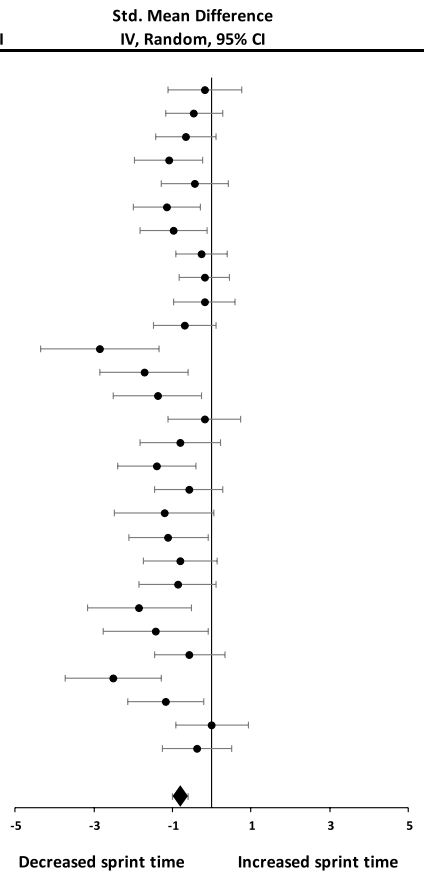

Fig. 3 Effect of resisted sprint training on early acceleration sprint performance. Circles represent the standardised mean difference (SMD) for each study group. Diamonds represent the pooled SMD for all study groups combined. *CI* confidence interval, *G1* group 1, *G2* group 2, *G3* group 3, *IV* independent variable, *SD* standard deviation

Study or Subgroup	RST Mean	SD	N	URS Mean	SD	N	Weight	Std. Mean Difference IV, Random, 95% CI
Cahill et al. [37] (G1)	-0.04	0.12	15	-0.04	0.13	12	11.30%	0.00 [-0.76, 0.76]
Cahill et al. [37] (G2)	-0.02	0.11	14	-0.04	0.13	12	10.90%	0.16 [-0.61, 0.93]
Cahill et al. [37] (G3)	0.00	0.09	12	-0.04	0.13	12	10.00%	0.35 [-0.46, 1.15]
Carlos-Vivas et al. [43] (G1)	-0.02	0.07	11	-0.02	0.08	12	9.70%	0.00 [-0.82, 0.82]
Carlos-Vivas et al. [43] (G2)	-0.02	0.03	13	-0.02	0.08	12	10.60%	0.00 [-0.78, 0.78]
Carlos-Vivas et al. [43] (G3)	-0.02	0.04	12	-0.02	0.08	12	10.20%	0.00 [-0.80, 0.80]
Gil et al. [44]	-0.01	0.15	9	0.01	0.17	9	7.60%	-0.12 [-1.04, 0.81]
McMorrow et al. [47]	0.00	0.09	6	0.02	0.04	7	5.40%	-0.28 [-1.37, 0.82]
Spinks et al. [67]	-0.05	0.05	10	-0.04	0.05	10	8.40%	-0.19 [-1.07, 0.69]
Total (95% CI)			**102**			**98**	**100.0%**	**0.01 [-0.26, 0.29]**

Heterogeneity: Tau² = 0.00; Chi² = 1.35, df = 10 (P = 0.99); I² = 0%
Test for overall effect: Z = 0.10 (P = 0.92)

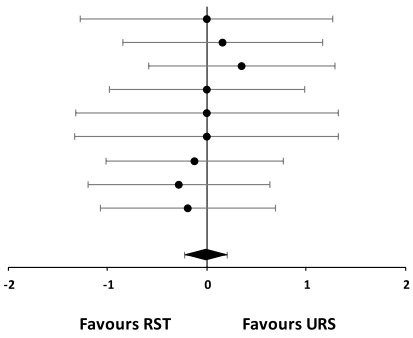

Fig. 4 Comparison of resisted sprint training (RST) and unresisted sprinting (URS) for improving late acceleration sprint performance. Circles represent the standardised mean difference (SMD) for each study group. Diamonds represent the pooled SMD for all study groups combined. *CI* confidence interval, *G1* group 1, *G2* group 2, *G3* group 3, *IV* independent variable, *SD* standard deviation

while also demonstrating small but significant effects for LA, but no improvement in MV performance. Additionally, while external load and loading methods did not appear to affect the outcome of RST interventions, interestingly, light and very heavy loads were similarly effective for developing the EA sprint phase.

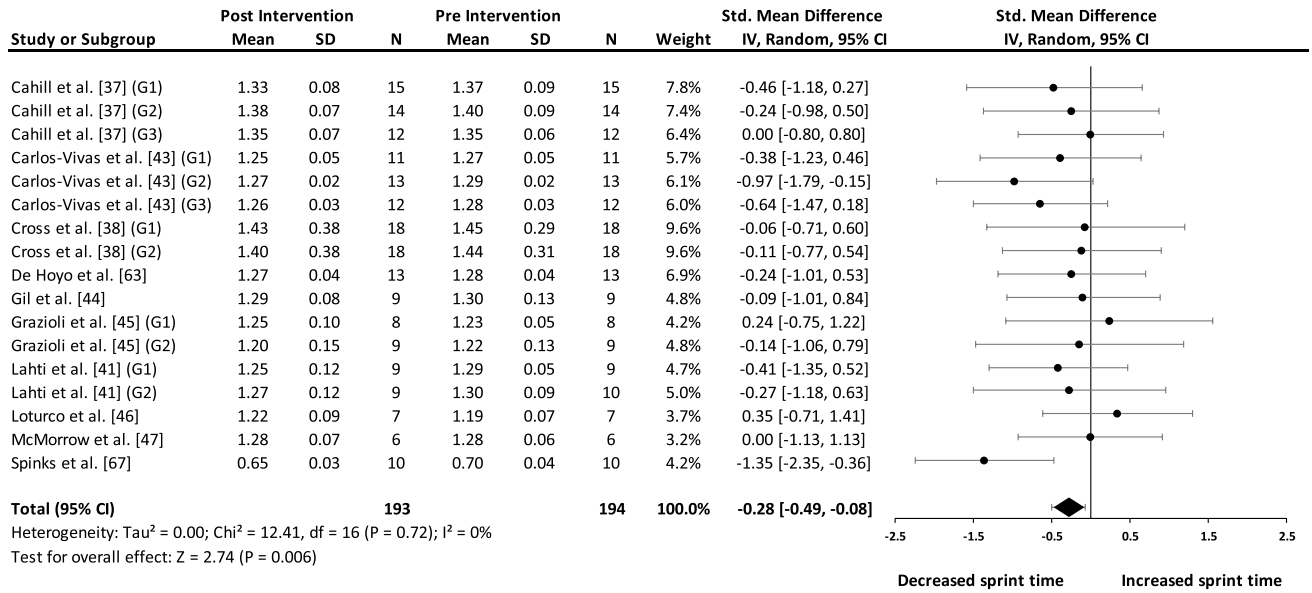

Fig. 5 Effect of resisted sprint training on late acceleration sprint performance. Circles represent the standardised mean difference (SMD) for each study group. Diamonds represent the pooled SMD for all study groups combined. *CI* confidence interval, *G1* group 1, *G2* group 2, *G3* group 3, *IV* independent variable, *SD* standard deviation

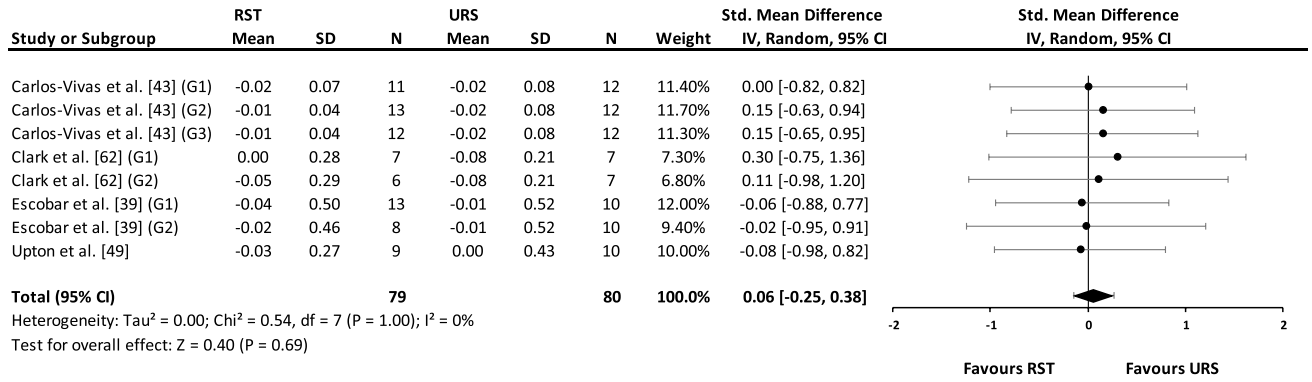

Fig. 6 Comparison of resisted sprint training (RST) and unresisted sprinting (URS) for improving maximum velocity sprint performance. Circles represent the standardised mean difference (SMD) for each study group. Diamonds represent the pooled SMD for all study groups combined. *IV* independent variable, *CI* confidence interval, *G1* group 1, *G2* group 2, *G3* group 3, *SD* standard deviation

4.1 Acceleration

No differences were detected between RST and URS for the development of EA and LA performance (Figs. 2, 4). Twenty-one out of 25 study groups in the current analysis demonstrated a mean decrease in sprint time following both RST and URS, indicating that both methods appear to improve, or at least maintain, acceleration performance. Although it is not possible to conclude this without the presence of a true control, this is supported by our secondary analysis that demonstrates that RST significantly improves both EA and LA. These findings agree with previous work [26, 28, 29] and suggest that both RST and URS may be viewed as specific and effective training methods for developing the acceleration performance of FITS players.

It is well established that acceleration is highly determined by the capacity to exert large propulsive horizontal force during the stance phase [71–74], which has been shown to be more important than the capacity to produce greater resultant force [10, 75]. Horizontally loaded RST modalities allow the player to adopt a forward oriented body position, indicative of accelerative sprinting, which aids the player in directing most of the force produced in a horizontal direction [20, 76], and thus possibly explains

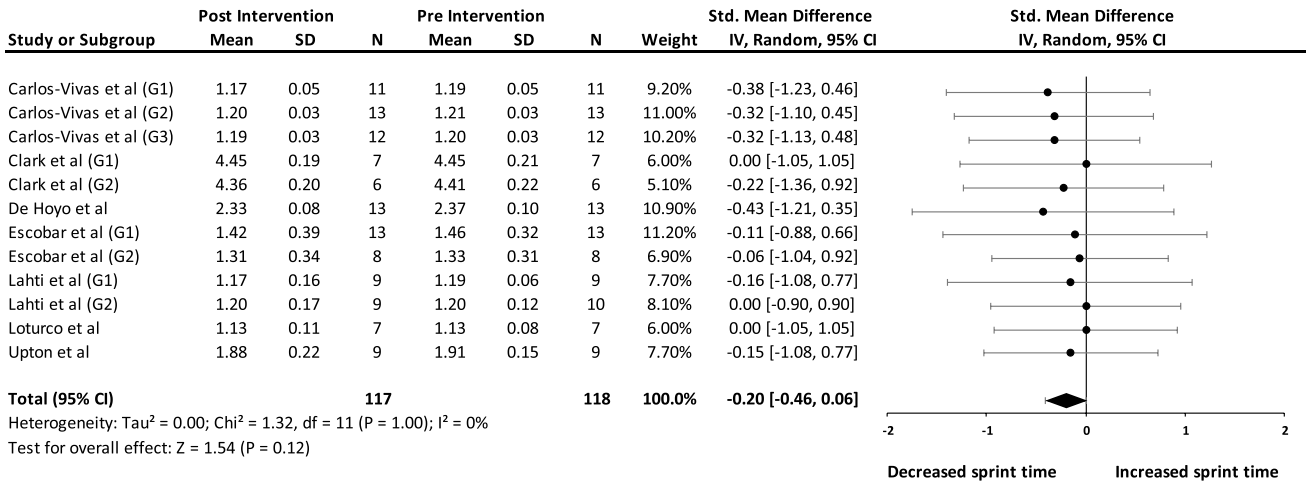

Fig. 7 Effect of resisted sprint training on maximum velocity sprint performance. Circles represents the standardised mean difference (SMD) for each study group. Diamonds represent the pooled SMD for all study groups combined. *CI* confidence interval, *G1* group 1, *G2* group 2, *G3* group 3, *IV* independent variable, *SD* standard deviation

why horizontally loaded RST modalities appear effective for improving acceleration performance [21, 37, 39–41]. Of the 16 study groups included in the EA phase analysis (Fig. 2), 13 used horizontally loaded modalities (i.e. sled [$n = 11$], Vertimax [$n = 1$], 1080 Sprint [$n =$])) [21, 37, 39, 43, 44, 47, 49, 63, 66–68], one included a combination of horizontal and vertically loaded modalities [43], while just two exclusively used vertically loaded modalities [43, 48] (Table S3 of the ESM). This facilitation of overload in a more forward-oriented position by horizontal RST devices not only increases horizontal force production and the technical ability to apply force but also enhances players' rate of force development [21, 41]. Thus, it appears that RST is an effective sprint training method for the FITS player given the overall reliance on acceleration during games [3, 77, 78].

While short-distance sprints are a common occurrence in FITS, a large variability in sprint distances performed by FITS players [77–79] means the LA phase is important if game or positional demands regularly expose players to longer sprint distances and MV sprinting. Furthermore, FITS players often begin sprinting from a variety of preceding movements [77], meaning they have already reached a considerable running speed by the time they initiate their sprint, and therefore likely often complete sprints whilst displaying LA characteristics. As athletes increase sprinting velocity and transition from EA to MV, they gradually adopt a more upright trunk position to that displayed in EA [80], which coincides with a gradual decrease in the ratio of force (horizontal force:resultant ground reaction force) [10] and an increase in braking and vertical force production [73, 81]. Despite these kinetic and kinematic changes, the goal for the athlete remains the same regardless of the sprint phase, i.e. produce as much horizontal force as possible [31]. Late acceleration phase performance requires force to be produced across shorter ground contacts at higher velocities relative to EA [82], and therefore RST may be less specific for improving LA than EA given that it requires longer ground contacts to optimise impulses [83, 84]. However, our results demonstrated no difference between RST and URS for improving LA performance (Fig. 4), and a small effect for RST (Fig. 5).

Sprint phase adaptations in response to RST are likely velocity, load and distance dependent [28, 50, 74, 85]. It has been suggested that heavier loads may be more beneficial for improving EA, while lighter loads (<10% V_{dec}) may be more effective for increasing the LA and MV phases [28]. Because of the inability of the %BM method to cater for differences in factors such as individual force–velocity and training characteristics, surface friction and body composition [28, 86, 87], and to provide more conclusive and transferable findings regarding RST load, our analysis was conducted using only studies which used the $\%V_{dec}$ method. Interestingly, and in contrast to the aforementioned research [28], we found moderate effects for both light (SMD − 0.69) and very heavy (SMD − 1.01) loads for improving EA performance (Table S5 of the ESM). Unfortunately, the absence of an appropriate number of study groups in the moderate and heavy load categories excluded these categories from analysis. This contrast in findings may be due to the greater number of studies included in this review utilising very heavy loads [37–41], in addition to the use of meta-analysis in comparison to the previous narrative approach. Unresisted EA typically displays the lowest velocity, longest ground contact times and greatest horizontal forces, relative to other sprint phases [75, 88], and thus displays mechanical characteristics more like those observed during RST [22, 27].

This may explain why RST appears to be most effective for improving EA relative to other sprint phases and may partly explain why loads at either end of the spectrum were found effective during this phase only.

Our findings appear to support the arguments presented by Zabaloy et al. [18] who questioned the need to use heavy and very heavy loads because of concerns around multiple factors such as disruptions to sprint technique and prolonged ground contact times. Large acute changes to URS kinematics and kinetics may lead to undesirable transfer to URS (e.g. increased trunk lean) [89]. However, recent work by Lahti et al. [41] demonstrate no long-term changes to EA or upright sprint kinematics following 9 weeks of very heavy RST. Despite this, limitations reported by the authors suggest these results should perhaps be interpreted with caution, and thus it is apparent that more research is needed in this area [18].

Light loads do not substantially interfere with sprint mechanics [83, 90] and thus may allow FITS players to better replicate LA phase characteristics and maximise transfer [90, 91]. Moderate and heavy loads (20% and 30% V_{dec}) require a significant increase in forward trunk lean to overcome these loads, suggesting that when sprinting with a more upright sprint posture as seen in LA and MV, perhaps only lighter loads or URS are suitable [44, 49, 67]. Our subgroup analysis somewhat supports this theory with light loads displaying the largest effect on LA phase sprint performance (SMD − 0.44). However, it should be noted that no study prescribed load with the specific intent of improving LA phase performance, and this should be considered by future research.

When interpreting our results, it is important to consider the method of load prescription utilised. Our analysis found both the $\%V_{dec}$ and %BM methods were effective for increasing EA performance, while only the $\%V_{dec}$ method was effective during LA (Table S6 of the ESM). It is well established that the $\%V_{dec}$ is superior to %BM when the goal is to prescribe load relative to the individual's force–velocity characteristics [28, 92–94], with numerous limitations to the %BM method reported [28, 86]. However, for some FITS teams, use of the $\%V_{dec}$ method is not a practical or feasible option. Prescribing light %BM loads [18] may be a possible solution for practitioners to deliver a more uniform training stimulus when using the %BM method, as %BM does not appear to reflect $\%V_{dec}$ at heavier loads [94]. Finally, despite the clear advantages of using the $\%V_{dec}$ method over the %BM method, no study has considered the reassessment of load velocity relationships during long-duration RST interventions. For example, during an 8- to 12-week training study, it is possible that an initially prescribed $\%V_{dec}$ load may fluctuate, increasing or decreasing depending on potential moderating factors such as fatigue, training history/level and individual adaptation [95–97].

4.2 Maximum Velocity

A between-group analysis found no differences between RST and URS for improving the MV phase (Fig. 6), while a within-group analysis detected no change in MV performance following RST (Fig. 7), the latter of which agrees with previous findings [26, 29]. Despite the prominence of shorter distance sprints within FITS [78, 98, 99], there is a strong relationship between MV and short sprint performance [30, 72, 100, 101] and players with a higher MV may have an advantage over players with a lower MV when performing shorter sprints [30]. Furthermore, given the range of maximum sprint distances recorded within FITS (e.g. 30–50 m [99, 102]), coupled with the knowledge that FITS players can achieve a high percentage of their MV at short distances [30, 103], it is reasonable to suggest that game/positional demands will present opportunities for players to reach MV and may therefore be just as important as acceleration within FITS.

Given the postural changes that accompany the attainment of MV such as an upright body position, increased MV performance is associated with the production of greater vertical forces during the early stance phase across very short ground contact times [104], in turn triggering a greater reliance on elastic properties and stiffness [73, 104–107]. This increase in vertical force application is required to achieve sufficient flight times to reposition the limbs for the following ground contact [108]. However, faster athletes produce greater horizontal force at any given velocity compared with slower athletes [31, 75], thus highlighting the importance of this component alongside vertical force during the MV phase.

Horizontal RST modalities are known to alter players' force orientation and allow greater horizontal force production [27]. Furthermore, the load velocity relationship during RST with sleds has been shown to be linear [93, 109], indicating that perhaps light loads are most appropriate for maintaining key MV sprinting characteristics and improving this sprint phase [28, 38, 43, 49, 63]. However, because of our external load analysis only including $\%V_{dec}$ studies (four study groups), all of which used very heavy loads, few conclusions could be made regarding the effect of external loads during the MV phase. Regardless, MV performance is related to the technical ability to apply ground reaction force more horizontally [10, 72, 75], and therefore the combination of horizontal modalities and light loads would appear useful for teaching FITS players how to generate more horizontal force from the more upright sprinting posture adopted during MV.

Acute application of heavier RST loads significantly alters MV phase characteristics in comparison to lighter loads [84, 90], for example, increased forward lean [84,

90], decreased flight time [84] and prolonged ground contact times [22], with experimental studies involving FITS players also showing reductions in MV [21, 37]. Thus, while these heavier loads will prioritise horizontal force production at lower velocities, this does not comply with key MV characteristics and possibly inhibits the important contribution of the stretch–shortening cycle and stiffness during this sprint phase [26, 84, 105, 110]. Therefore, it may be more appropriate for coaches to develop the MV phase using URS until such time adaptations cease, and perhaps only then should an overload of MV sprinting via RST (using light loads [43, 49, 63]) be considered necessary.

Alternatively, research [18, 40, 41] suggests that combining URS alongside heavy and very heavy loaded RST may attenuate potential negative effects to MV demonstrated in previous research [21, 37]. Indeed, both methods offer unique benefits and therefore their inclusion within speed training programmes should likely not be viewed as a 'one or the other' debate. For example, given that FITS players must develop multiple physical qualities concurrently during training and competitive periods [35, 111], it is possible that a similar approach could be taken when developing sprint performance, i.e. multiple methods used to simultaneously develop distinctive sprint specific components such as horizontal force production [21], power [93] and velocity [37]. Additionally, given the ability of RST to simultaneously develop these sprint specific qualities, it may be an effective method for FITS coaches to increase training efficiency during short, congested training/competitive periods, replacing gym-based strength and/or power training while maintaining a velocity dominant stimulus with URS. However, further work is required to investigate the potential of combined sprint training practices, specifically regarding the magnitude of the RST loads being used and the RST/URS volumes being implemented.

Finally, of the eight study groups included in the primary MV analysis, just two solely used weighted vests applying light and moderate loads, respectively (Carlos-Vivas et al. [43] group 1, Clark et al. [62] group 2) (Fig. 6), with just one group displaying a small increase in MV [43]. Applying a vertically oriented overload to the sprinting action in the form of weighted vests has been shown to induce acute increases in peak vertical ground reaction force and ground contact time during MV (using an approximate relative load of 22% BM), while simultaneously decreasing velocity, step length, flight time and horizontal force [24]. It is thought that weighted vest sprinting may be more effective for increasing MV performance than acceleration due to the increased contribution of vertical force during this phase; however, studies have reported similar improvements to all sprint phases [43, 48, 62]. The primary goal of sprinting is to move horizontally as fast as possible, and although vertical force production plays a significant role in achieving higher top speeds, it is crucial to ensure optimal force distribution between both horizontal and vertical components to avoid potential negative effects to MV phase performance (e.g., increased flight times and reduced step frequency [91]). In this regard, outside of its possible benefit of improving the contribution of the stretch–shortening cycle [20, 24] and stiffness, the use of vertically oriented RST devices for increasing MV phase performance remains unclear given the lack of studies investigating its longitudinal effects on sprint performance [23].

4.3 Limitations

The use of two different loading methods in the included studies, and the inconsistencies associated with the %BM method, make interpreting and comparing the results of RST studies difficult. A limited number of study groups eligible for the subgroup analysis of external load inhibited our analysis of this important moderator variable, particularly within the LA and MV phase analyses where conclusions could only be met regarding the effects of light and very heavy loads. Therefore, we recommend future RST studies use the $\%V_{dec}$ method to prescribe load, while also providing more detail on the training practices/history (particularly sprint and strength training) of participant groups. Because of multiple studies lacking a URS control group, a lower number of study groups were included in the main analysis compared with the secondary analysis, and therefore we recommend future RST studies include a URS control group. The inclusion of a greater number of studies utilising heavier loading during the LA and MV phase in our within-group analyses (Figs. 5, 7) may mean that our findings do not truly reflect the specific ability of RST to improve these phases and thus future work should aim to prescribe RST with the deliberate intent of enhancing LA and MV performance, an approach likely utilising lighter loads. Finally, because of the concurrent training practices within FITS, participants completed additional training alongside their sprint training protocols, for example, sports training, resistance training and plyometrics (Table S4 of the ESM). Therefore, as is typically associated with completing experimental research within FITS settings, determining with certainty the true cause of effect is challenging despite the assumption that the only difference between groups was the type of sprint training protocol completed. While it is essential to acknowledge this as a limitation, it is also important to be aware of the ecological validity associated with the inclusion of these applied practices within FITS training interventions.

5 Conclusions

Our review demonstrates that RST induces similar improvements to EA, LA, and MV phase sprint performance when compared with URS. Secondly, our analysis indicates that RST

is most effective for improving EA followed by LA, with no improvements found during the MV phase. Both light and very heavy loads were effective for increasing EA performance, while greater benefits were associated with using the $\%V_{dec}$ loading method. In summary, RST is an effective and versatile sprint training method available to FITS coaches aiming to improve the sprint performance of their players.

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Declarations

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Conflict of interest Cormac Ward, Ciarán Ó Catháin, Niamh Ní Chéilleachair, Shaun Grassick and David T. Kelly have no conflicts of interest that are directly relevant to the content of this article.

Consent to participate Not applicable.

Consent for publication Not applicable.

Availability of data and material Datasets analysed and/or produced during this study are available from the corresponding author.

Code availability Not applicable.

Authors' contributions CW, DTK, COC and NNC contributed to the conception and design of the study. CW and SG conducted the literature search and data extraction. CW drafted the manuscript and DTK, COC and NNC contributed to the editing and revision of the manuscript. All authors read and approved the final manuscript prior to submission.

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